

Unordered Selections

Class Session Goals:

- 1 Introduce Notation for Unordered selections and derive a formula
- 2 Present the Idea of a Combinatorial Proof
- 3 Work on applying the multiplication principle
- 4 Present connection of 'Choose' numbers and Binomial coefficients

Order and Repetition

How many ways are there to make k choices from a list of n possibilities? In order to make this a question that we can answer, we must answer the following two fundamental yes/no questions of Chapter 1.

Repetition - Are we able to repeat choices?

Order - Does the order in which we make the choices matter?

For a given counting question, different answers are possible. Goal of this chapter to create general formulas for each of the four situations.

Some Answers

Order Matters, Repetition not allowed: The number of ways to make k choices from n where repetition is not allowed and order matters is

$$n \cdot (n - 1) \cdot (n - 2) \cdot (n - 3) \cdots (n - k + 1) = \frac{n!}{(n-k)!}.$$

Order Matters, Repetition Allowed: The number of ways to make k choices from n where repetition is allowed and order matters is n^k .

Today's question: How many ways are there to make k choices from n possibilities where repetition is not allowed and order does not matter?

Binomial Coefficients

Give the answer a name: Let $\binom{n}{k}$ be the number of ways to make k choices from n objects where order does not matter and repetition is not allowed. Sometimes books use the notation $C(n, k)$ for this.

Some Examples

$$\binom{n}{1} =$$

Binomial Coefficients

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Some Examples

$$\binom{n}{1} = n$$

$$\binom{n}{n} =$$

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$$\binom{n}{1} = n$$

$$\binom{n}{n} = 1$$

$$\binom{n}{0} =$$

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Some Examples

$$\binom{n}{1} = n$$

$$\binom{n}{n} = 1$$

$$\binom{n}{0} = 1$$

$$\binom{n}{2} =$$

Binomial Coefficients

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Some Examples

$$\binom{n}{1} = n$$

$$\binom{n}{n} = 1$$

$$\binom{n}{0} = 1$$

$$\binom{n}{2} = \frac{n(n-1)}{2}$$

Our goal is general formula for $\binom{n}{k}$

Activity

Down Right paths on an $n \times n$ grid.

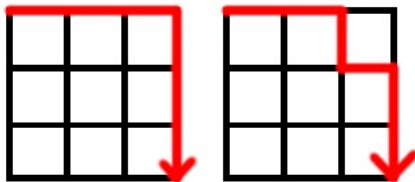


Figure: Two down-right paths on a 3×3 grid.

Combinatorial Proofs

Idea Count a set S in two different ways to get to different counting formulas. The two formulas must be equal because they are counting the same set.

Combinatorial Proof Example How many ways are there to select a k person team from n tryouts and select a captain for the team?

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$$k \cdot \binom{n}{k} = n \cdot \binom{n-1}{k-1}$$

We can use this to create a recursive algorithm to calculate $\binom{n}{k}$.

Unordered Selection Formula

The number of ways to make k choices from n objects where order does not matter and repetition is not allowed is

$$\binom{n}{k} = \frac{n!}{k! \cdot (n - k)!}$$

The derivation of this is our first example of a direct proof in this course.

Revisit Degenerate Example

How many balanced bit strings of length 4 (or length n while were at it)?

Two stage process: first place the 0s, second place 1s.

The number of **balanced bit string** of length $2n$ is $\binom{2n}{n}$.

The number of **balanced ternary strings** of length $3n$ is $\binom{3n}{n} \binom{2n}{n}$.

Balanced ternary strings of length $3n$ have n 0s, n 1s, and n 2s.